

2025

CAB230 Assignment 2 Client Side

CAB230

Movies API – Client Side Application

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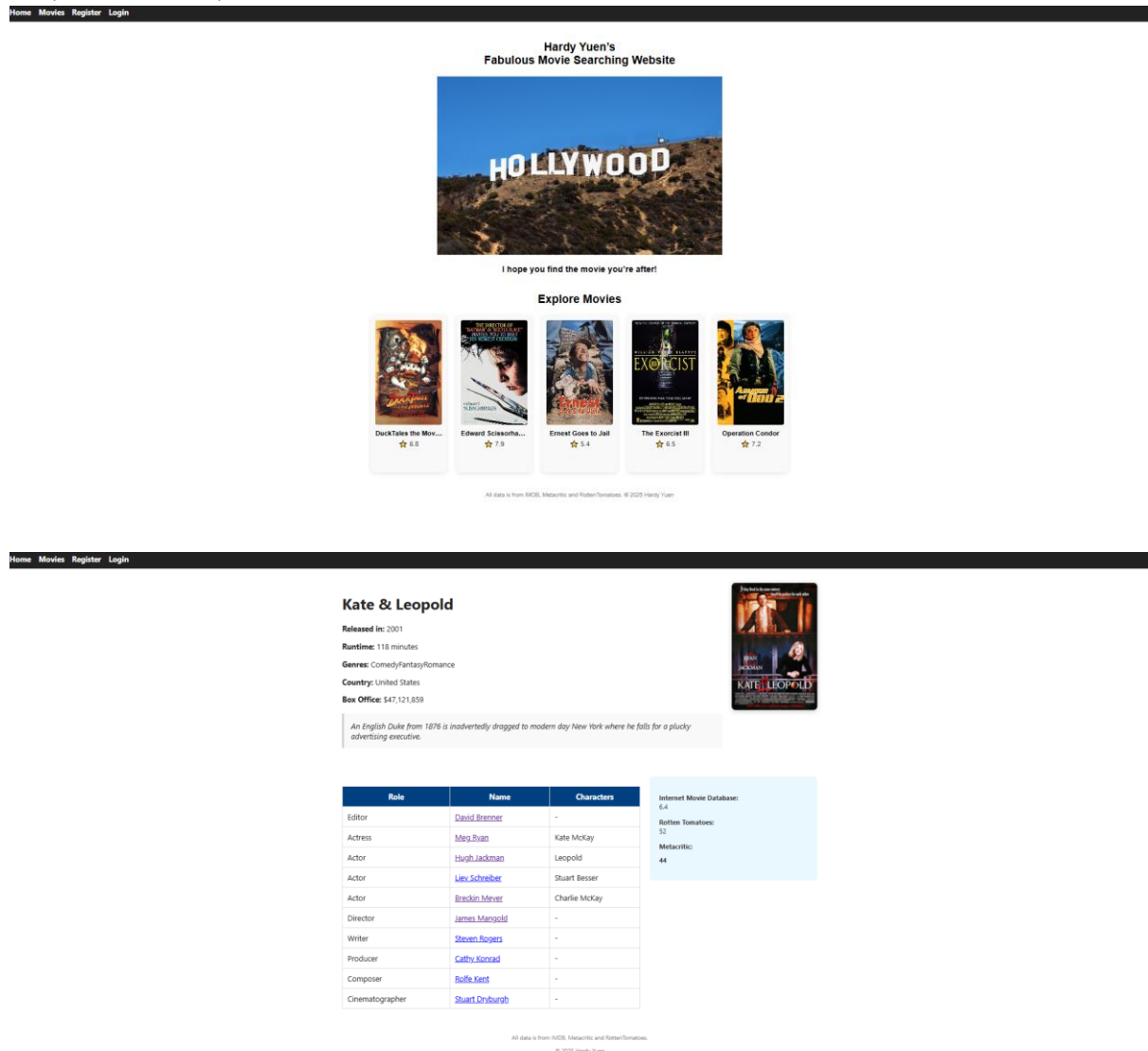
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This template is adapted from one created for a more elaborate application. The original author spends most of his professional life talking to clients and producing architecture and services reports. You may find this a bit more elaborate than you are used to, but it is there to help you get a better mark

This report should be around 10 pages or so including screenshots – there is no formal page limit, and the length will depend a lot on the number of screen shots, but you won't get any extra marks for a really long report.

Introduction

Purpose & description



This web application is a movie exploration platform that allows users to search, browse, and view detailed information about films. It connects to a live movie API to retrieve real-time data including titles, ratings, posters, and cast details. Users can register and log in to access protected features such as viewing individual actor and crew profiles along with their film histories.

The homepage features a rotating carousel of movie posters that updates every ten seconds, offering a simple way to discover random titles. The Movies page includes a search function with filters for title and year, and supports infinite scrolling to load large sets of results efficiently.

Selecting a movie opens a detail view that displays information such as the release year, runtime, genres, box office performance, and a full cast list. Clicking on a cast member leads to their profile page, which shows the roles they've played and includes a bar chart visualising the IMDb ratings of those films.

Routing is handled using React Router, and secure access is managed with bearer and refresh tokens to maintain user sessions. Additional features such as fallback images for missing posters, interactive charts, and structured pagination help create a more complete and user-friendly experience.

Completeness and Limitations

This app meets most of the expectations for a **Grade 6 level**. It supports user registration and login, uses bearer and refresh tokens to keep users logged in, and allows authenticated access to protected endpoints. Navigation is managed through React Router and all forms are fully controlled. The movie search includes pagination through infinite scrolling, and search filtering by title and year is functional.

The Person page includes a chart showing IMDb rating distribution using Chart.js, aligning with the requirement to visualize data meaningfully. While the app does not use AG Grid specifically, a clean custom table was built to present data clearly. The overall UI is minimal and easy to use, though styling could be further improved to enhance polish. All core features work smoothly and without major bugs.

Use of End Points

[*/movies/search*](#)

This endpoint powers the Movies page. Users can search by title and filter by year. The results load in batches using infinite scroll and are displayed in a paginated table.

[*/movies/data/{imdbID}*](#)

This is used both on the Landing page (for pulling full poster and rating data) and the Movie Details page. It shows metadata like genres, runtime, box office, and cast.

[*/people/{id}*](#)

This endpoint is protected and only accessible after login. It fetches detailed info about a person, including their roles in movies and IMDb ratings. A bar chart visualizes the distribution of those ratings.

[*/user/register*](#)

Used on the Register page. New users can sign up using email and password, with basic client-side validation.

[*/user/login*](#)

Used on the Login page. Logs the user in, stores bearer and refresh tokens in localStorage, and updates app context.

[*/user/refresh*](#)

Called automatically to refresh the session when the bearer token expires. This helps avoid making the user log in repeatedly.

[*/user/logout*](#)

This endpoint is now fully implemented. When a user clicks the logout button, a POST request is sent to `/user/logout` using the stored refresh token. After the logout request completes, both the bearer and refresh tokens are removed from localStorage, and the user is redirected to the login page..

Modules Used

Chart.js / react-chartjs-2

Used to render the bar chart on the Person Details page, showing IMDb rating distributions in a clean and interactive way.

<https://www.chartjs.org/>

React Router DOM

Handles routing between pages, including parameter-based navigation like /movies/:imdbID and /people/:personID.

<https://reactrouter.com/en/main>

React Context API

Used through a custom AuthContext to manage user authentication state across the app.

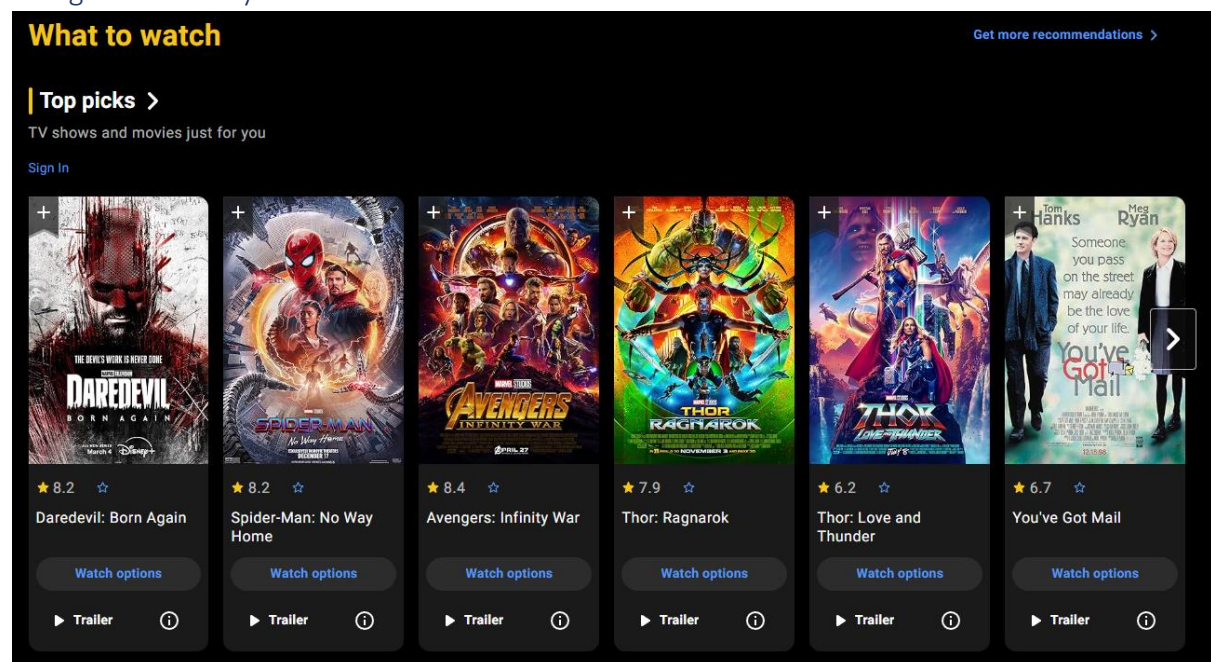
Intersection Observer

Used with a ref to implement infinite scrolling on the Movies page, triggering more data fetches as the user scrolls.

https://developer.mozilla.org/en-US/docs/Web/API/Intersection_Observer_API

Application Design

Navigation and Layout



I didn't really go through a formal design process for this project. Instead, I looked at sites like IMDb and some streaming platforms to get a feel for how movies are usually laid out. I didn't create any mockups or sketches, but I tried to copy the things that made sense and kept the structure simple so users could move around easily.

The navigation is straightforward. At the top there's a menu with links to Home, Movies, and either login or logout depending on whether the user is signed in. Once you're logged in, you land on a homepage that has a rotating set of movie posters which changes every 10 seconds. That was a fun way to make the page feel a bit more dynamic, kind of like Netflix's carousel style.

The Movies page lets you search by title and filter by year. Results load automatically as you scroll, and if you click on a title you get a full breakdown on the Movie Details page including runtime, genres, box office numbers and cast. From there, clicking on any actor or crew member takes you to their Person page which shows the movies they worked on and includes a chart that shows their IMDb rating distribution.

I built most of the layout by just trying things out in the browser and adjusting based on what looked clean and easy to use. I didn't want anything too fancy, just something that made sense and worked smoothly.

Usability and Quality of Design

The overall layout of the application is clean and logically structured. Each page is focused on a specific task, such as exploring movies, viewing details, or checking cast and crew information. The design keeps things simple and avoids clutter, which helps users find what they need without being overwhelmed. The movie search table presents data in a clear and readable format, and the rotating carousel on the homepage adds some visual interest.

Navigation is straightforward and consistent across the application. The top menu provides easy access to the Home and Movies pages and adjusts based on the user's login status. Page transitions are smooth, and links behave as expected. Users can move between movies and people easily, and the flow of the app matches common patterns found on similar websites.

That said, there are definitely areas that could be improved. The visual design is minimal and doesn't include much styling beyond the basics. While fonts and sizes are consistent, the app lacks a strong visual identity and could benefit from more attention to colour, spacing, and layout polish. The UI is functional, but some parts may feel too plain or unfinished, especially compared to commercial movie platforms.

Another limitation is mobile responsiveness. The app works well on desktop, but it hasn't been fully optimised for smaller screens. Some elements may appear off-centre or cramped when viewed on a phone or tablet. This could be improved by applying more responsive CSS rules or using a modern UI library.

In terms of usability, the app meets expectations. It is easy to navigate, visually consistent, and performs as intended. If more time were available, improvements could be made to enhance the overall design quality, accessibility, and mobile experience.

Accessibility

The application meets some basic accessibility guidelines, but there are still several areas where improvements could be made to fully align with W3C's Priority 1 requirements.

For non-text elements, I have included alt attributes for images across the site. Movie posters on the landing and detail pages have descriptive alt text that includes the movie title, which helps provide context for screen reader users. However, fallback images or icons may need better descriptions to be fully compliant. The site does not include audio or video content, so no captions or transcripts were necessary.

Most of the content is accessible without relying solely on colour. For example, genre tags and ratings are displayed as plain text, not colour-coded indicators. This helps ensure that users who are colour-blind or using monochrome displays can still interpret the information. That said, some parts of the layout (such as genre badges) could benefit from stronger visual contrast to improve readability.

If the CSS styles were removed, the raw HTML structure of the app would still be understandable. Headings, paragraphs, and labels are used appropriately, so the flow of content should still make sense without styling. However, visual spacing is doing a lot of work right now, so without CSS, some sections may look cramped or difficult to follow.

Dynamic content, such as the movie carousel or search results, is updated in a way that visually reflects changes, but it does not currently include ARIA roles or live region announcements for screen readers. This would be important to consider for a more accessible implementation of dynamic content.

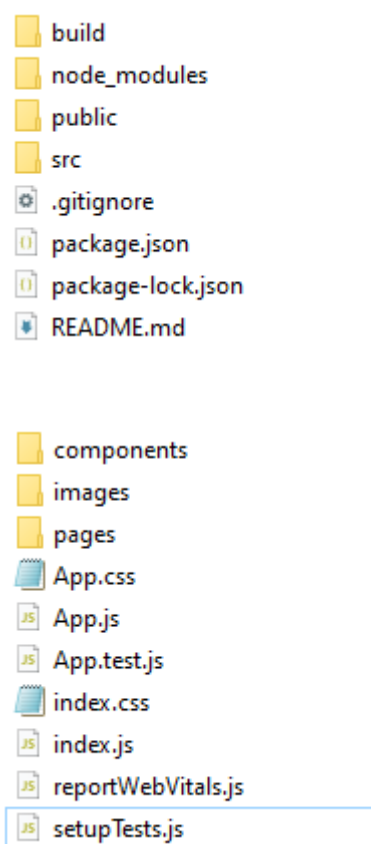
There is no flashing or flickering content in the app, so it is unlikely to trigger issues for users with photosensitivity. Language used throughout the site is simple and conversational, which supports readability for a wide audience, including users with cognitive disabilities or limited English proficiency.

For tables, the structure is quite basic. I use `<thead>` and `<tbody>` properly, and the headings are clear, but I have not explicitly defined ARIA roles or scope attributes to indicate relationships between headers and cells. This could be added to improve compatibility with screen readers.

In summary, while the app meets a few core accessibility standards such as proper alt text, clear headings, and readable layout without relying on colour, it lacks deeper features like ARIA roles, semantic table markup, or screen reader support for dynamic updates. If I had more time, I would focus on improving keyboard navigation, contrast ratios, and adding accessibility metadata to make the app more inclusive.

Technical Description

Architecture



This application follows a typical React project structure, with all main functionality living inside the `src` folder. The top-level project contains standard files like `package.json`, `.gitignore`, and folders such as `node_modules` and `public`, which are used for dependency management and static files respectively.

Inside the `src` folder, responsibilities are clearly split into logical subdirectories:

- **components/**: This folder holds reusable UI components like navigation bars, forms, or table rows. Keeping these separate makes the UI modular and easier to maintain or reuse.
- **images/**: This directory stores static assets used in the interface, such as the landing page background image or fallback posters.

- **pages/**: Each major route or screen in the app has its own file here—like LandingPage, LoginPage, MoviesPage, and so on. Each page component is responsible for rendering a full view and orchestrating its logic and subcomponents.

Files like App.js serve as the entry point for routing and layout. index.js is the root rendering file, bootstrapping the app into the DOM. Supporting files such as reportWebVitals.js and setupTests.js are part of the React app template, supporting performance tracking and unit testing.

This structure was chosen to clearly separate the **presentation layer (components)**, **static assets (images)**, and **page-level views (pages)**. This modular setup improves clarity, keeps responsibilities organized, and helps scale or debug the app more easily.

All routing and token-based access control are handled through the main App.js and component state/context. Pages like MoviesPage and PersonDetails fetch live API data and display it using paginated tables and charts, while state and UI logic remain localized within each component.

Test plan

Task	Expected Outcome	Result	Screenshot Ref
Register with valid email and password	Account is created and success message is shown	PASS	01
Register with mismatched passwords	Error message is shown: “Passwords do not match”	PASS	02
Login with correct credentials	User is logged in and redirected to homepage	PASS	03
Login with incorrect password	Error message is shown: “Login failed”	PASS	04
Search for existing movie title	Matching movies are shown in the table	PASS	05
Search with empty field	All results are shown (default behaviour)	PASS	06
Search for movie with a rare or invalid title	No results displayed or message shown	PASS	07
Navigate to movie details page	Movie information, cast, and ratings are displayed	PASS	08
Navigate to person page from cast list	List of roles and IMDb rating chart is shown	PASS	09
Scroll down movies page (infinite scroll test)	More results load automatically	PASS	10
Access person page without login	Error message is shown: “You must be logged in”	PASS	11

Difficulties / Exclusions / unresolved & persistent errors /

During development, there were a few challenges that took time to work through. One of the main issues was getting the API data to load correctly, especially when chaining multiple asynchronous fetches. At first, some requests would fail silently or return undefined values, which turned out to be due to incorrect handling of promises and missing error checks. Once I added better error handling and filtered out null responses, the data loaded much more reliably.

Styling also took longer than expected. Positioning elements like the movie cards on the landing page and aligning content across different screen sizes was tricky without using a full design framework. I had to manually adjust spacing, font sizes, and layout to make everything look consistent, and this process involved a lot of trial and error.

One of the biggest unresolved problems is related to authentication tokens. After logging in, if I immediately click into the movie page, the data sometimes fails to load. But if I wait one or two minutes and refresh, the information appears. I suspect this has something to do with the timing of token storage or the bearer token not being available quickly enough after login. Despite multiple attempts to debug this issue, it still occasionally occurs and remains unresolved.

Lastly, I had planned to polish some UI components further and possibly implement sorting for the movie table, but ran out of time. While the core features work, some aspects like responsiveness on mobile and advanced filtering are still quite basic.

Extensions (Optional)

There are several directions this app could grow in the future to improve both functionality and user experience:

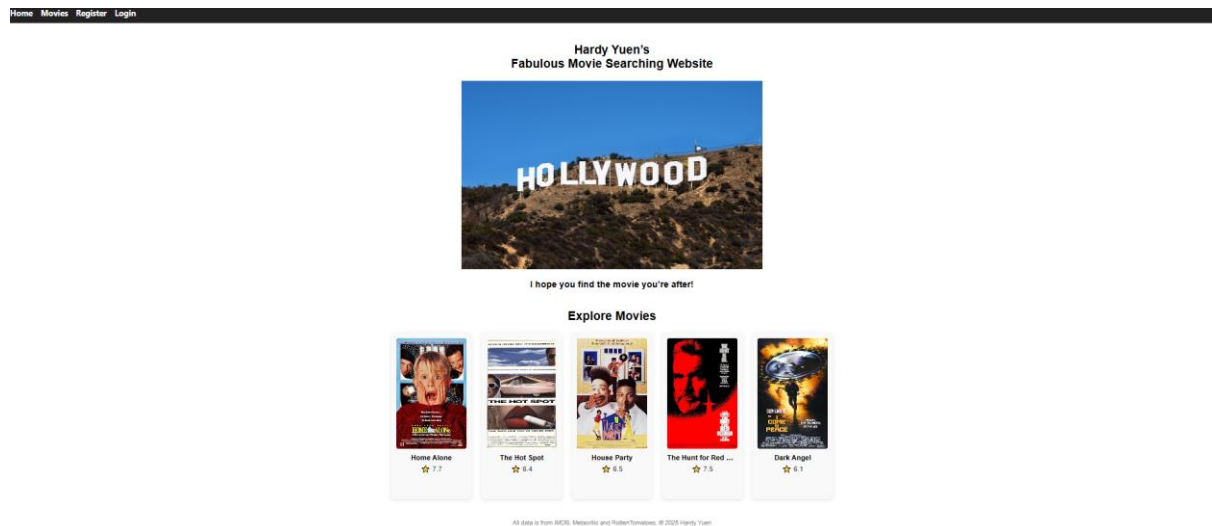
1. **Watchlist or Favourites:** Allow logged-in users to bookmark or save movies and people to a personal list. This would require an extra backend endpoint and user-linked database support.
2. **Dark Mode Toggle:** Adding a dark mode theme switcher using local storage or context would improve accessibility and user comfort for night-time use.
3. **Trailer Integration:** Embedding trailers from YouTube or other sources directly into the movie detail page could make the experience more interactive.
4. **Advanced Search Filters:** Right now, users can only search by title and year. Filters for genre, country, or rating range would make the search more powerful.
5. **Improved Mobile Responsiveness:** The design is usable on mobile but could be refined further with mobile-first layout optimizations.
6. **Authentication Enhancements:** Support for OAuth (Google/Facebook login) or multi-factor authentication could make the login process more secure and user-friendly.

These features would enhance usability, increase engagement, and better showcase the capabilities of the movie dataset. They also provide good technical challenges for future development.

User guide

1. Landing Page

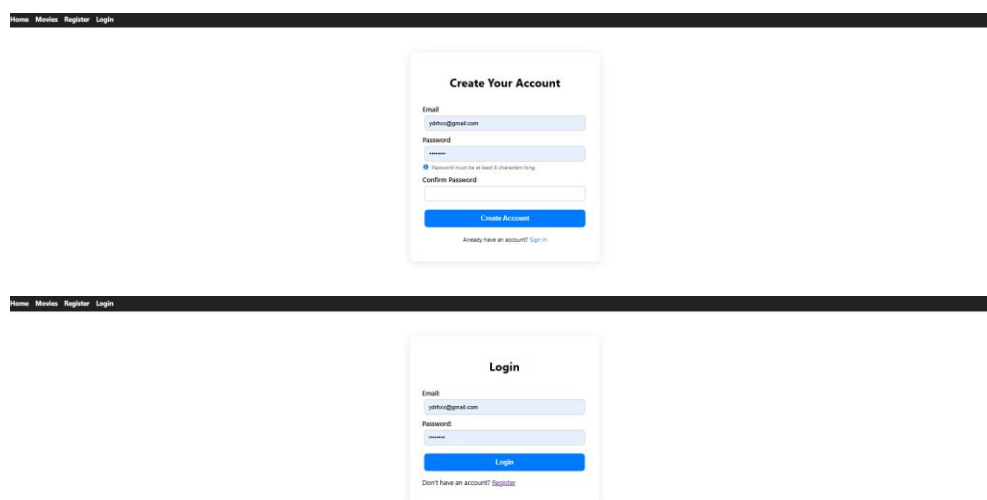
When you first visit the site, you'll be greeted by a welcoming landing page with a rotating carousel of movie cards. This carousel updates every 10 seconds and displays the title, poster, and IMDb rating for each movie.



2. Registration and Login

To access full features like detailed person pages and authenticated movie data, you must register and log in.

- Click on **Register** in the top navigation bar.
- Fill in your email, password (minimum 8 characters), and confirm your password.
- After successful registration, you'll be redirected to the login page.
- On the **Login** screen, enter your email and password to access the app.



The login process also stores your session using tokens so that you stay authenticated when browsing protected content.

3. Browse Movies

Click **Movies** from the navbar to access the searchable movie table. Here you can:

- Filter movies by title or release year.
- Change the number of results shown per page (10, 50, 100).
- Scroll down to trigger automatic loading of more results via infinite scrolling.

Movies

Search movies...

Any year

100

Search

Showing 100 of 12184 results

Title	Year	IMDb	Rotten Tomatoes	Metacritic	Classification
Kate & Leopold	2001	6.4	52	44	PG-13
The Other Side of the Wind	2018	6.7	83	78	R
A Tale of Springtime	1990	7.1	86	-	PG
Moon 44	1990	5	0	39	R
The Match Factory Girl	1990	7.5	-	-	Not Rated
Three Men and a Little Lady	1990	5.4	40	51	PG
Boiling Point	1990	6.7	94	-	Not Rated
The Adventures of Ford Fairlane	1990	6.4	25	24	R
Agneseth	1990	7.6	-	-	Not Rated
Air America	1990	5.8	13	33	R
Alice	1990	6.6	71	67	PG-13
An Angel at My Table	1990	7.5	95	79	R
Another 48 Hrs.	1990	5.9	20	23	R
Arachnophobia	1990	6.5	83	67	PG-13
Avolon	1990	7.2	86	68	PG
Awakenings	1990	7.8	86	74	PG-13
Back to the Future Part II	1990	7.4	89	55	PG
Bad Influence	1990	6.2	65	63	R
Basket Case 2	1990	5.4	71	-	R
Bird on a Wire	1990	6	26	36	PG-13
Blue Steel	1990	5.7	75	54	R
The Bonfire of the Vanities	1990	5.6	15	27	R
Boyz n the Ring	1990	6.2	37	54	R

Each row includes IMDb, Rotten Tomatoes, and Metacritic ratings, along with classification info. Click a movie title to view its full details.

4. Movie Details

The **Movie Details** page displays:

- Title, year, runtime, genres, box office, and plot
- A full cast and crew table with clickable names
- Rating breakdown from IMDb, Rotten Tomatoes, and Metacritic

The Other Side of the Wind

Released in: 2018

Runtime: 122 minutes

Genre: Drama

Country: France, Iran, United States

Box Office: 5

At a media overflow party to celebrate his 70th birthday and screen his avant-garde film-in-progress, a legendary but jaded Hollywood director is faced both with voracious fans and unsettling questions about what became of his last...

THE OTHER SIDE OF THE WIND

2018

Role	Name	Characters
Editor	Bob Muzewski	-
Actor	John Huston	Jake Hannaford
Actress	Oja Kodar	The Actress
Actor	Peter Bogdanovich	Brooks Otterlake
Actress	Susan Strasberg	Julie Rich
Director	Orson Welles	-
Producer	Frank Marshall	-
Producer	Ellie Jan Pymaza	-
Composer	Michel Legrand	-
Cinematographer	Gary Graver	-

Internet Movie Database: 6.7

Rotten Tomatoes: 83

Metacritic: 78

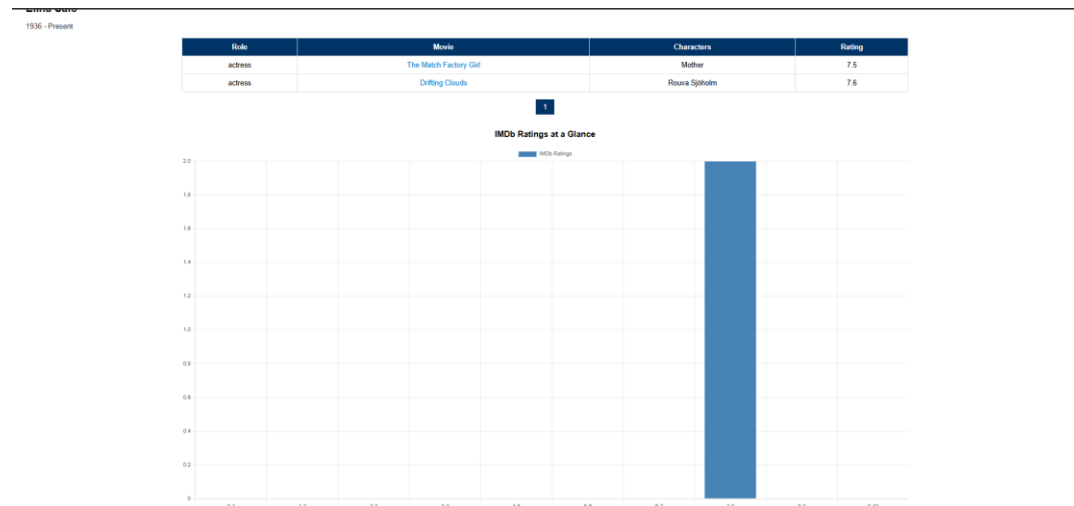
All data is from IMDb, Metacritic and RottenTomatoes.

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5. Person Details

When clicking a cast or crew member, you're taken to a **Person Details** page, which shows:

- Their lifespan (birth/death years)
- A list of all their credited roles across movies
- A chart showing IMDb rating distribution for their roles



References

Wikipedia contributors. (n.d.). *Hollywood Sign*. Wikipedia.

https://en.wikipedia.org/wiki/Hollywood_Sign

Appendices as you require them

Ref 1:

Create Your Account

Email

Password

Password must be at least 8 characters long.

Confirm Password

[Create Account](#)

Registration successful!

Already have an account? [Sign in](#)

Ref 2:

Create Your Account

Email

ydrhxx@gmail.com

Password

Password must be at least 8 characters long.

Confirm Password

**

Create Account

Passwords do not match.

Already have an account? [Sign in](#)

Ref 3:

Home

Movies

Welcome, ydrhxx@gmail.com

Logout

Hardy Yuen's
Fabulous Movie Searching Website



I hope you find the movie you're after!

Explore Movies



Kate & Leopold
★ 4.4



The Other Side of L...
★ 6.7



A Tale of Springtime
★ 7.1



Moon 44
★ 5



The Match Factory...
★ 7.5

All data is from IMDb, Metacritic and Rotten Tomatoes. © 2020 Hardy Yuen

Update password?

Username

ydrhxx@gmail.com

Password

**

Update password

No thanks

You can use saved passwords on any device. They're saved to [Google Password Manager](#) for mdeatr1ght1226@gmail.com.

Ref 4:

Login

Email:

ydrhxx@gmail.com

Password:

Login

Incorrect email or password

Don't have an account? [Register](#)

Ref 5:

star

Any year

100

Search

Title
Star Trek VI: The Undiscovered Country
Star Trek: Generations
Stargate
The Star Maker
Lone Star
Star Trek: First Contact
Starship Troopers
Star Trek: Insurrection
Star Wars: Episode I - The Phantom Menace
Star Wars: Episode II - Attack of the Clones
Star Wars: Episode III - Revenge of the Sith
Superstar
Barking at the Stars
Rock Star
Star Trek: Nemesis
Stark Raving Mad
Dickie Roberts: Former Child Star
Starsky & Hutch
Starter for 10
Stardust
Star Trek
Bright Star
Like Stars on Earth
Star Wars: The Clone Wars

Ref 6:

Search results

Any year

100

Search

Showing 100 of 12754 results					
Title	Year	IMDb	Number of Votes	Reviews	Classification
Wish I Could	2001	6.4	10	46	PG-13
The Other Side of the Wind	2015	6.7	83	76	R
A Year of Springtime	1990	7.1	99	-	PG
Mean St	1990	5	5	39	R
The World's Fastest Girl	1990	7.5	-	-	Not Rated
Three Men and a Cradle	1990	5.4	46	51	PG
Baking Point	1990	6.7	94	-	Not Rated
The Adventures of Ford Fairbank	1990	6.4	35	24	R
Agony	1990	7.6	-	-	Not Rated
Big Business	1990	5.9	13	33	R
John	1990	6.6	71	67	PG-13
An Angel at My Side	1990	7.5	95	79	R
Another at the	1990	6.9	36	29	R
Backstage	1990	6.5	93	67	PG-13
Avanti	1990	7.2	96	68	PG
Avantaggio	1990	7.8	85	74	PG-13
Back to the Future Part II	1990	7.8	98	96	PG
Bad Influence	1990	6.2	85	63	R

Ref 7:

and

Any year

100

Search

Showing 0 of 0 results				
Title	Year	IMDb	Number of Votes	Reviews

Ref 8:

The Other Side of the Wind

Released in: 2018

Runtime: 122 minutes

Genres: Drama

Country: France, Iran, United States

Box Office: \$

At a media-swamped party to celebrate his 70th birthday and screen his avant-garde film-in-progress, a legendary but jaded Hollywood director is faced both with voracious fans and unsettling questions about what became of his land...



Role	Name	Characters
Editor	Bob Murawski	-
Actor	John Huston	Jake Hannaford
Actress	Oja Kodar	The Actress
Actor	Peter Bogdanovich	Brooks Otterlake
Actress	Susan Strasberg	Julie Rich
Director	Orson Welles	-
Producer	Frank Marshall	-
Producer	Filip Jan Rymasz	-
Composer	Michel Legrand	-
Cinematographer	Gary Graver	-

Internet Movie Database: 6.7

Rotten Tomatoes: 83

Metacritic: 78

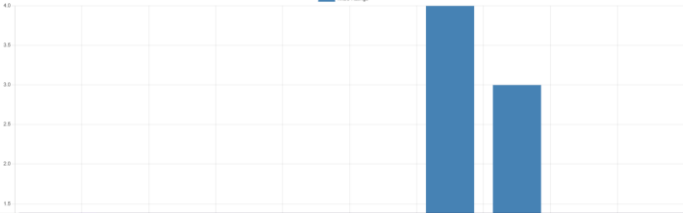
Ref 9:

Peter Bogdanovich

1939 - 2022

Role	Movie	Characters	Rating
actor	The Other Side of the Wind	Brooks Otterlake	6.7
director	Noten Off	-	7.5
director	The Thing Called Love	-	6.4
director	The Cat's Meow	-	6.4
director	She's Funny That Way	-	6.1
self	Hitchcock/Troubad	Self	7.3
self	They'll Love Me When I'm Dead	Self - Actor, Director	7.4

IMDb Ratings at a Glance



Ref 10:

The Doors	1991	7.2	57	62	R
Double Impact	1991	5.6	33	40	R
The Double Life of Véronique	1991	7.7	84	86	R
Drop Dead Fred	1991	5.9	11	25	PG-13
Dutch	1991	6.5	17	-	PG-13
Dying Young	1991	6	25	-	R
Enchanted April	1991	7.3	88	78	PG
Ernest Scared Stupid	1991	5.7	17	38	PG
Europa	1991	7.6	82	69	R
FOX2	1991	5.8	38	48	PG-13
Father of the Bride	1991	6.5	70	51	PG
The Fisher King	1991	7.5	84	61	R
Firting	1991	7.1	96	-	R
For the Boys	1991	6.3	44	-	R
Frankie and Johnny	1991	6.7	69	66	R
Freddie's Dead: The Final Nightmare	1991	4.7	23	39	R
Fried Green Tomatoes	1991	7.7	75	64	PG-13

Ref:11

You must be logged in to view this page.

Go to Login